

TECHNICAL GENERAL

BASICS

General Definitions:

- **Mass (kilogram-Kg):** The quantity of matter in a body and is a measure of how difficult it is to start or stop. Mass influence time and/or distance required to change the direction of a body.
- **Weight (Newton – N):** The force due to the gravity. ($F = m \times g$), $g = 9.81 \text{ m/s}^2$
- **Density (kg/m^3):** Matter (Mass) in a body per unit volume. ($\rho = m/V$)
- **Velocity (metre per second – m/s):** Rate of change of displacement.
- **Acceleration (metre per second square – m/s^2):** Rate of change of velocity
- **Momentum (Mass x Velocity – kg m/s):** The quantity of motion possessed by a body.
- **Force (Newton – N):** A push or pull causing a change in direction or motion of body. $F = ma$; Units: N (Newton) or lb (Pound-force).
- **Centre of Gravity (CG):** Point through which the weight of an aircraft acts and around which aircraft rotates in flight.
- **Work (Joule – J):** Work is done on a body when it moves in the direction in which the force is acting. Work = Force x Distance. Units: J (Joules) or foot-pound
- **Power (Watt – W):** Power is rate of doing work. Units: (Joules/sec = watts.

$$\text{Power} = \frac{\text{Work}}{\text{time}} = \frac{\text{Force (N)} \times \text{Distance (m)}}{\text{time}}$$

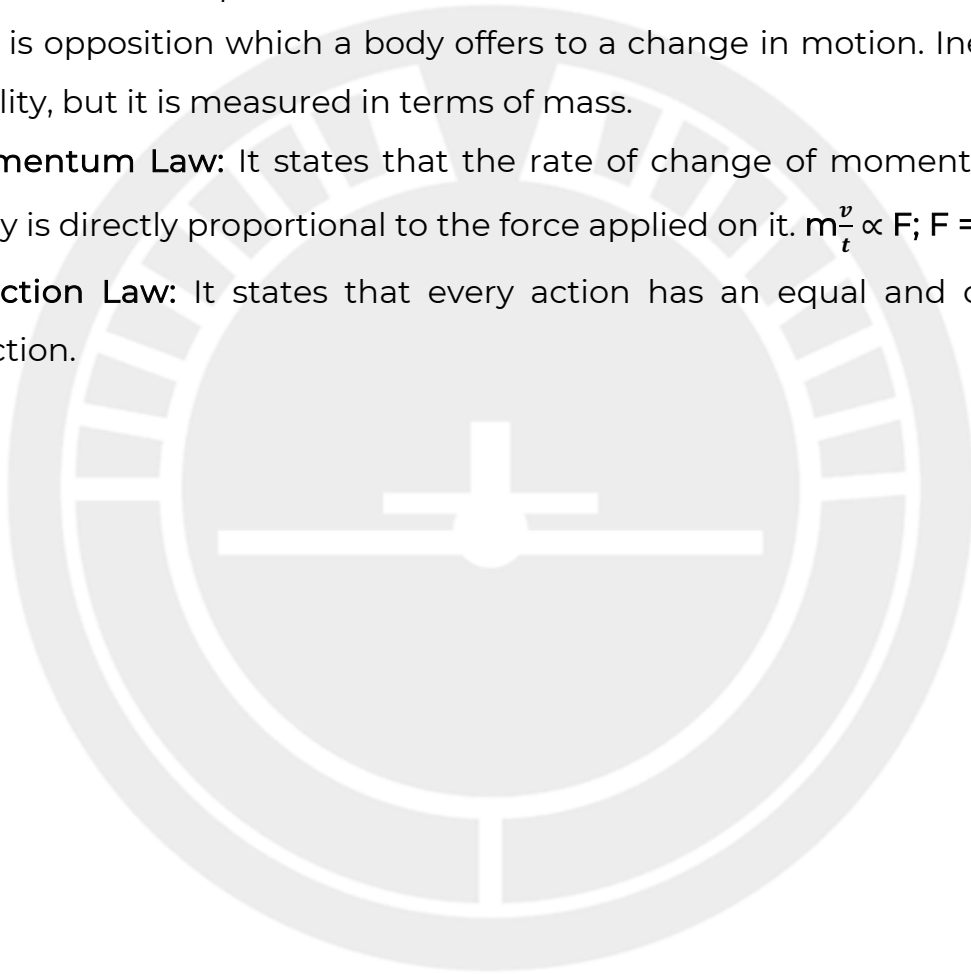
- **Energy (Joule – J):** The capacity to do work.
Law of conservation of energy: states that "*Energy can neither be created nor destroyed; only changes from one form to another.*"
- **Potential Energy (Joule – J):** Energy possessed by a body by virtue of its height. $\text{PE} = \text{mass} \times \text{gravity} \times \text{height}$
- **Kinetic Energy (Joule – J):** The energy possessed by mass because of its motion.

$$\text{KE} = \frac{1}{2}mV^2$$

- **Pressure (Pascal – Pa or N/m²):** Force acting per unit area. (*1 hp(hectopascal) = 1 mb(millibar) = 100 Pa*)
- **Frequency (Hertz – Hz):** Number of cycles per second / or Rotation per minute (rpm)

Newton's Laws of Motion:

- **Inertia Law:** It states that a body at rest or in uniform motion will continue to be in the same position unless and until an external force is acted upon it. It is opposition which a body offers to a change in motion. Inertia is a quality, but it is measured in terms of mass.
- **Momentum Law:** It states that the rate of change of momentum of a body is directly proportional to the force applied on it. $m \frac{v}{t} \propto F$; $F = ma$
- **Reaction Law:** It states that every action has an equal and opposite reaction.



Atmosphere

The most importance property of atmosphere is air density ("thickness" of air). A given mass flow will generate the required lift force, but a decrease in air density will reduce the mass flow thus the speed of the aircraft must be increased to maintain the mass flow and lift force at its required value.

Physical properties of air:

Air is a compressible fluid and is able to flow or change its shape when subjected to even minute pressure differences. Air will flow in the direction of lower pressure. Since air is compressible, lower layers contain much greater part of whole mass of the atmosphere.

International Standard Atmosphere (ISA):

- To enable accurate comparison of aircraft performance and calibration of pressure instruments, a 'standard' atmosphere has been adopted which represents mean or average properties of the atmosphere. Mean Sea Level values:

• Temperature	15°C
• Pressure	1013.25 hPa
• Density	1.225 kg/m ³

Temperature:

- Temperature in atm. is due to heat radiated from earth's surface.
- Temperature decreases with increasing altitude at a uniform lapse rate of 1.98°C per 1000 ft (approx. 2°C) from mean sea level up to a height of 36090 ft above which temperature remains constant at -56.5°C.

Static Pressure:

- It is the pressure acting on an object due to column of stationary air above that object. It is the result of the weight of the atmosphere pressing down on the air beneath.

- Unit – N/m^2 (1 hp (hectopascal) = 1 mb (millibar) = 100 Pa = 100 N/m^2).

Dynamic Pressure (N/m^2):

- When the velocity of any moving fluid or object is reduced due to any obstruction in its path, its kinetic energy is converted to pressure (total energy remains constant), known as dynamic pressure. Aircraft immersed in moving airflow will experience both static and dynamic pressure.
- Dynamic pressure (θ) = $\frac{1}{2} \rho V^2 \text{ N/m}^2$
- Total Pressure (Stagnation or Pitot Pressure) = Static Pressure + Dynamic Pressure

Air Density (kg/m^3):

- Density is mass per unit volume. Density varies with static pressure, temperature and humidity.
 - Temperature – increasing temperature decreases air density. $\rho \propto 1/T$
 - Static Pressure – decreasing static pressure decrease air density. $\rho \propto P$
 - Humidity – increasing humidity decrease air density. $\rho \propto 1/\text{Humidity}$

Note: (Density of water vapour is about 5/8 that of dry air)

- Density decreases with increasing altitude because of decreasing static pressure. However, with increasing altitude temperature also decreases, which would tend to increase density, but the effect of decreasing static pressure is dominant.

Basic Aerodynamics

Principle of Continuity:

- Fundamental law of the universe is **Energy and Mass can neither be created nor destroyed**, only changed from one form to another.
- The air mass flow, or mass per unit time, through the tube will be the product of the cross-sectional area (A), the airflow velocity (V) and the Air density (ρ).
- Mass flow will remain a constant value at all points along the tube, it is called Principle of Continuity (also known as *Constant Mass Flow Rate*).
The Equation of Continuity is:

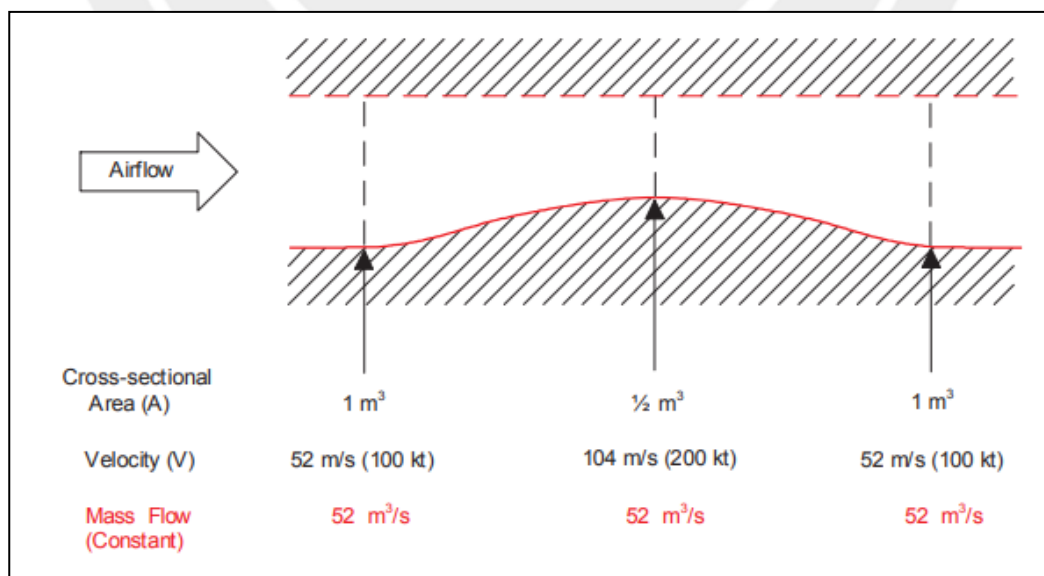
$$\frac{\text{mass}}{\text{time}} = \rho AV - \text{constant}$$

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 ; \text{ This is known as equation of Continuity.}$$

$$A \times V \times \rho = \text{Constant}$$

- At low subsonic speeds ($< M 0.4$) density changes will be insignificant and can be disregarded. The equation of continuity is simplified to:

$$A \times V = \text{constant}$$



Bernoulli's Principle of Continuity

Bernoulli's Theorem:

- “In the steady flow of an ideal fluid the sum of the pressure energy and the kinetic energy remains constant”. *Ideal fluid is incompressible and has no viscosity.*

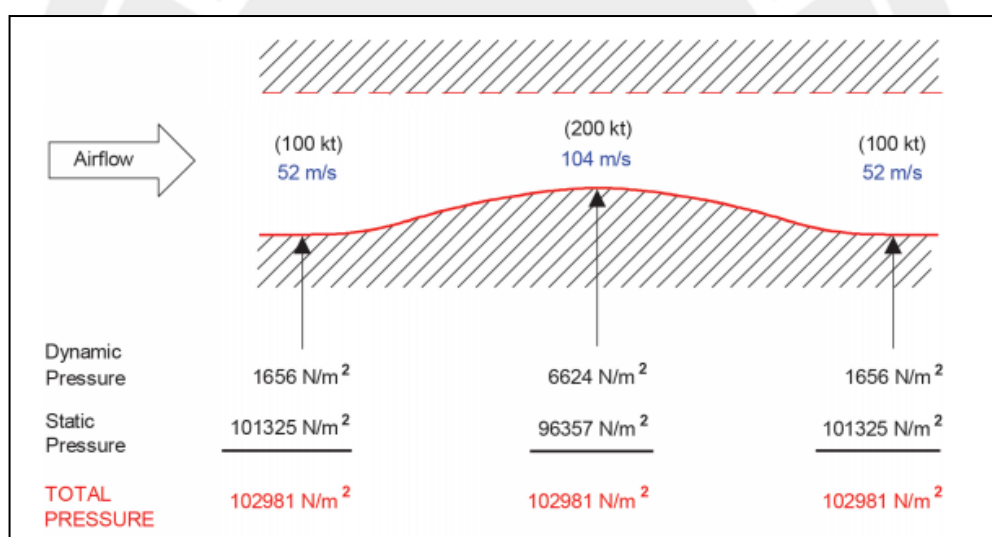
Pressure Energy + Kinetic Energy + Potential energy = Total energy (Constant)

- If height is constant, pressure energy is constant and Bernoulli's Equation is given as:

Pressure energy + Kinetic energy = Constant or:

$$p + \frac{1}{2} \rho V^2 = \text{Constant}$$

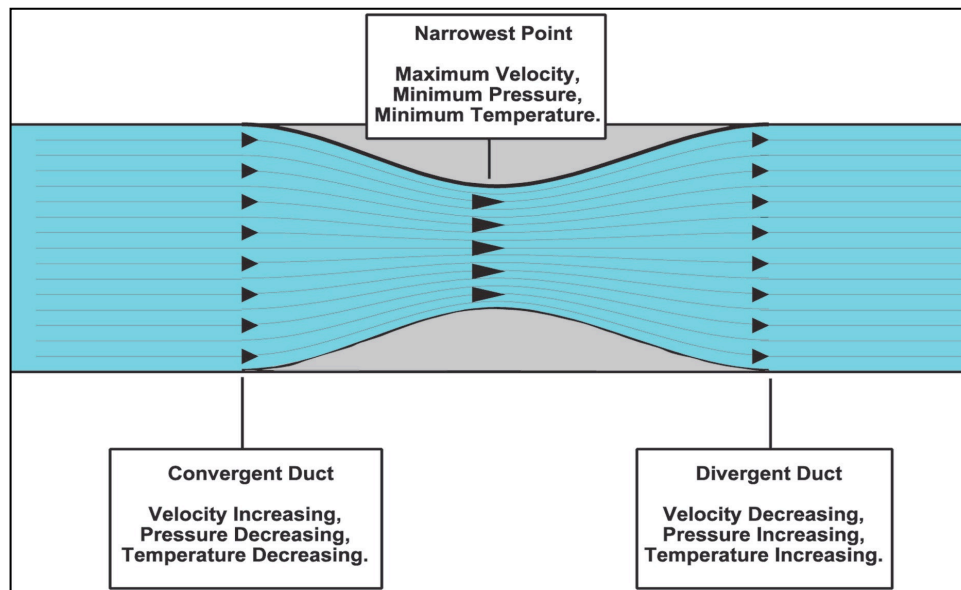
- Static Pressure + Dynamic Pressure = Total Pressure (constant). This total pressure can be called as Stagnation pressure or Pitot pressure.



Ducts:

There are 3 types of ducts:

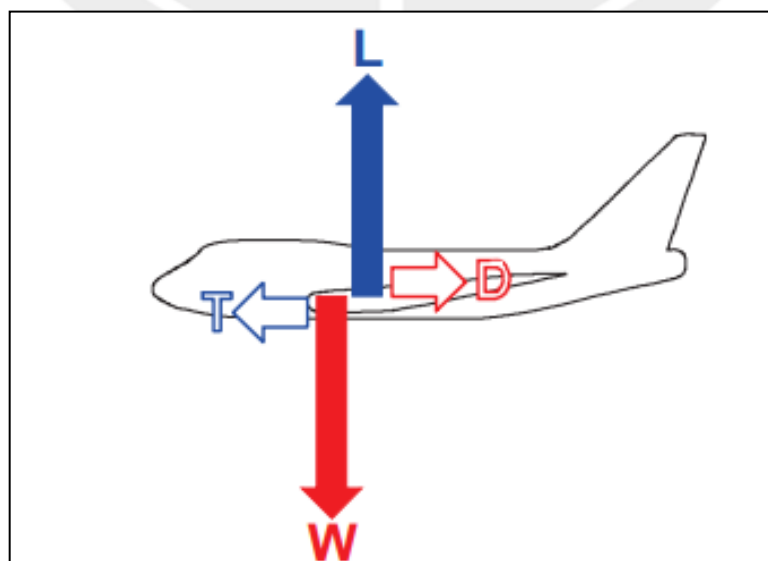
1. Convergent: A duct in which the cross-sectional area of the duct is decreasing.
2. Divergent: A duct in which the cross-sectional area of the duct is increasing.
3. Convergent-Divergent (C-D duct) or **Venturi**: It is a combination of convergent and divergent ducts.



Venturi Tube

Forces acting on an aircraft:

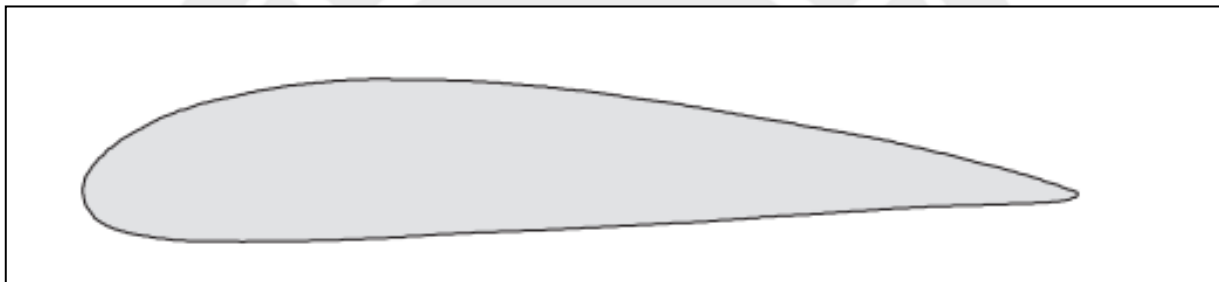
- Weight is force due to the acceleration of gravity acting upon it in vertically downward direction.
- Lift is the force produced by wings in upward direction which helps aircraft to take-off by defying gravity and sustain aircraft in flight.
- Drag is the force produced in direction opposite to aircraft's movement due to friction between aircraft and air molecules.
- Thrust is produced by engines which helps the aircraft to move in forward direction by countering drag.



- The greater the weight, the greater the lift requirement.
- The greater the drag, the greater the thrust required.

Aerofoil:

- Flow velocity is dependent on the shape of the object over which it flows. An increase in velocity will cause an increase in dynamic pressure and a decrease in static pressure.
- An aerodynamic force, called Lift, to oppose the weight of an aircraft can be generated by using a specially shaped body called an aerofoil.
- An aerofoil is 2-D profile of the wing, and is designed in aerodynamic shape to give maximum lift and minimum drag.



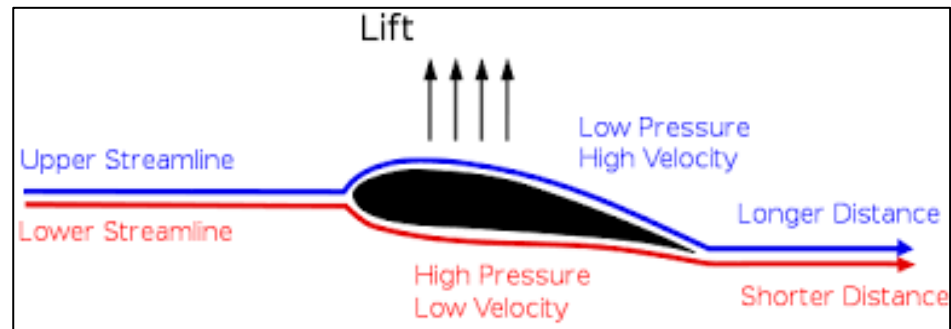
An aerofoil shape

1. **Aerofoil:** A shape capable of producing lift with relatively high efficiency.
2. **Leading edge (LE):** Leading most point of aerofoil.
3. **Trailing Edge (TE):** Trailing most point of aerofoil.
4. **Chord Line:** A straight line joining leading and trailing edge of an aerofoil.
5. **Angle of Attack:** (α or alpha) Angle between the chord line and the relative airflow.

Lift Generation:

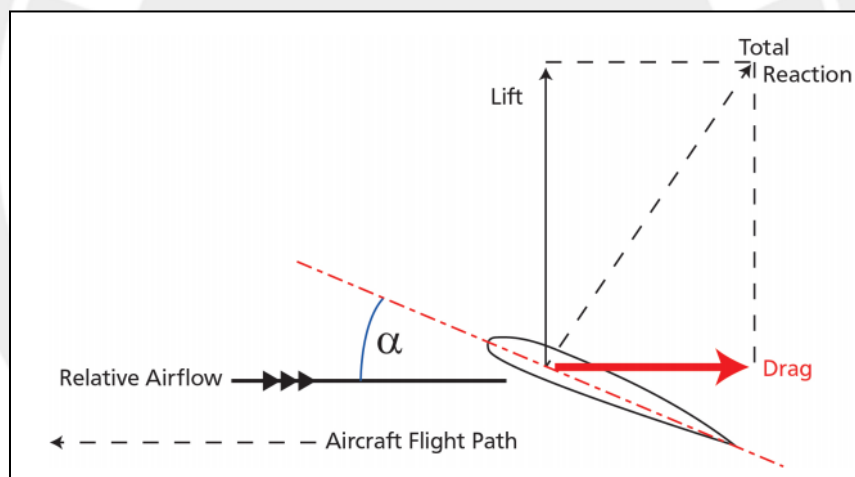
- **Wing Design** - Upper surface curved more compared to lower surface, causing air over upper surface to converge more compared to lower surface.

- Hence $\uparrow$ in velocity & $\downarrow$ in pressure over upper surface is more compared to lower surface, leading to **low pressure on upper & high pressure on lower surface**.
- Pressure difference b/w lower & upper surface creates a force in upper direction - called Lift.



Aerodynamic Forces:

- Forces produced due to movement of wing through the air – Lift & Drag.



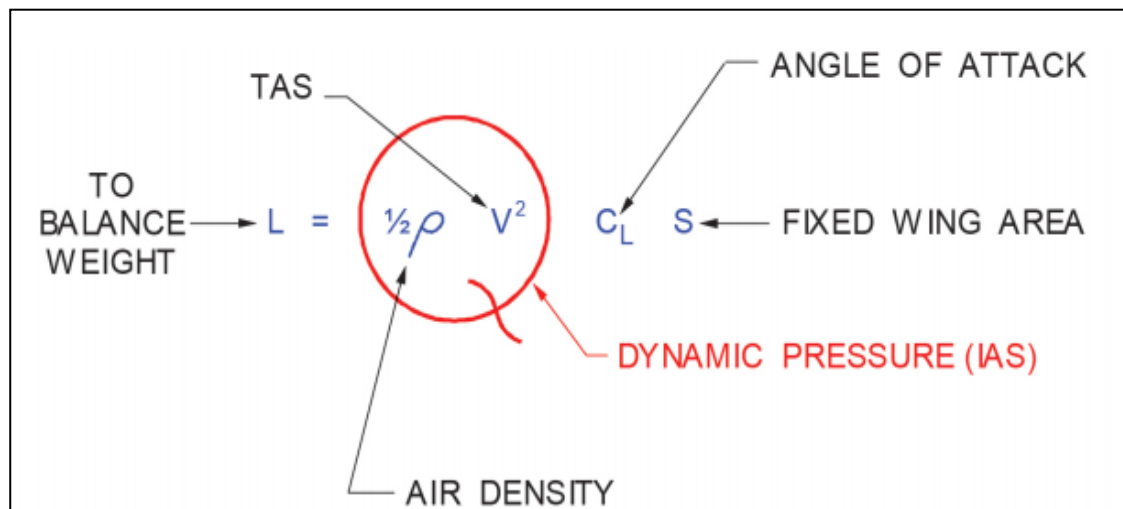
Angle of Attack: (α or alpha) Angle between the chord line and the relative airflow.

- **Total Reaction:** Resultant of all the aerodynamic forces acting on the aerofoil section.
- **Lift:** Aerodynamic force which acts at 90° to the relative airflow.
- **Drag:** Aerodynamic force which acts parallel to and in the same direction as the relative airflow or opposite to the aircraft flight path.

Basic Lift Equation:

- Aerodynamic Force (lift or drag) is given as: $F = Q C_F S$.
- Lift is net force generated normal (at 90°) to the relative airflow or flight path of the aircraft. The lift force is given as:

$$L = \frac{1}{2} \rho V^2 C_L S$$



Basic Drag Equation:

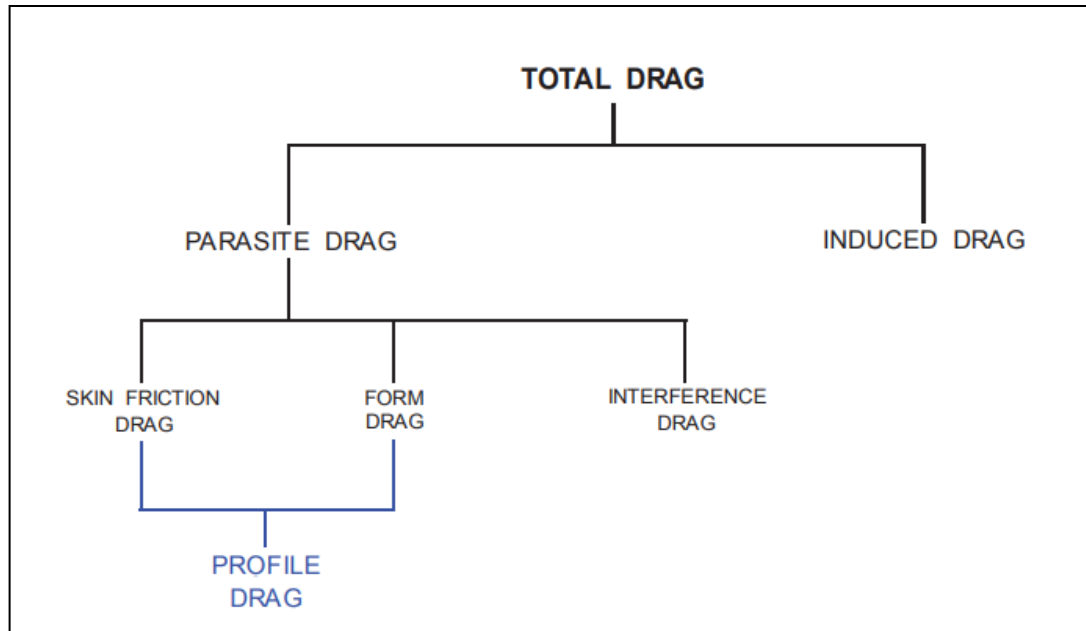
- Drag is the force which resists the forward motion of the aircraft. Drag acts parallel in direction of relative airflow (in the opposite direction to the flight path). The drag force is given as:

$$D = \frac{1}{2} \rho V^2 C_D S$$

- Every part of an aeroplane exposed to the airflow produces different types of resistance to forward motion which contribute to the Total Drag. Total Drag is sub-divided into two main types:
 - **Parasite Drag:** independent of lift generation, and
 - **Induced Drag:** result of lift generation.
- Parasite drag is further sub-divided into:
 - Skin Friction Drag
 - It is the drag due to friction between air molecules and surface of the aircraft.
 - Form (Pressure) Drag
 - The drag produced due to shape of the object.
 - The pressure differential between the leading edge stagnation point and the lower pressure at the trailing edge creates a force acting backward which is called '**form**' (pressure) drag.

- Interference Drag
 - It results from boundary layer 'interference' at wing/fuselage, wing/engine nacelle and other such junctions.

NOTE: Skin friction and Form drag are together known as Profile drag.



Wing Tip Vortices:

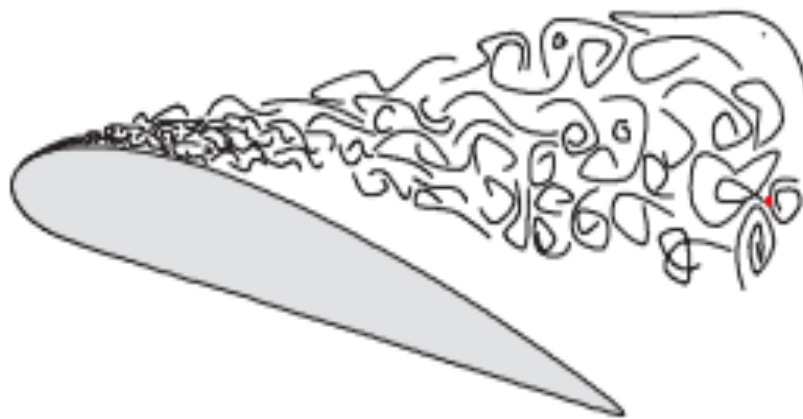
- Air flowing over the top surface of a wing is at a lower pressure & interact with high pressure air from bottom surface at the trailing edge and the wing tips.
- The greater the pressure difference between the top and bottom of the wing, the stronger the vortices around the wingtip.
- The larger the “generating” aircraft relative to the aircraft encountering the wake turbulence, the greater the hazard.

Induced Drag:

- Formation of these wing tip vortices leads to additional drag – called Induced drag. The stronger the vortices, the greater the induced drag.
- **Winglets & Sharklets:** A surface placed at the wing tip will reduce the formation of tip vortices by restricting airflow from lower surface to upper surface.

Stall:

- Stalling is a hazardous manoeuvre involving loss of height and loss of control.
- At low AoA no flow separation occurs. As angle of attack increases, adverse pressure gradient increases, reducing the kinetic energy, and boundary layer will begin to separate from surface at trailing edge.
- At angles of attack higher than approx. 16° , airflow will separate from the surface, causing low pressure area on the top of the section to suddenly collapse. This condition is known as **stall**.



High Lift Devices:

- Aircraft are fitted with high lift devices
 - to reduce the take-off and landing speeds and
 - to reduce take-off and landing distances
 - to permits operation at greater weights from given runway lengths.
- A flap is a hinged portion of the trailing or leading edge which can be deflected downwards to increase camber.
- Smaller aircrafts have trailing edge flaps only, but larger aircrafts usually have both leading edge (slats) and trailing edge (flaps) devices.

Landing Gear:

Tricycle (Nose Wheel)	Tail Dragger (Tail Wheel)
Most aircraft use this layout where 2 main wheels are placed after CG and	This layout has two main wheels and a tail wheel at back.

one nose wheel. Supports 90% of aircraft weight & initial landing shocks.

Nose wheel is used for steering on ground and leveling the aircraft.

It has danger of tipping it over onto its nose while taxiing in a strong wind and also of ground looping.



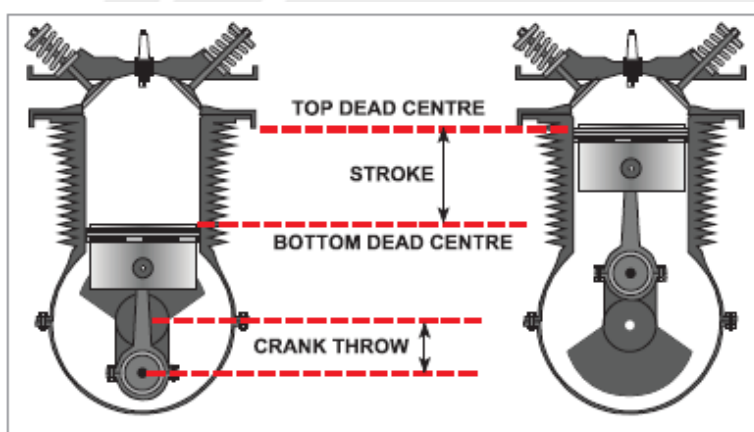
PISTON ENGINE

Gas Laws:

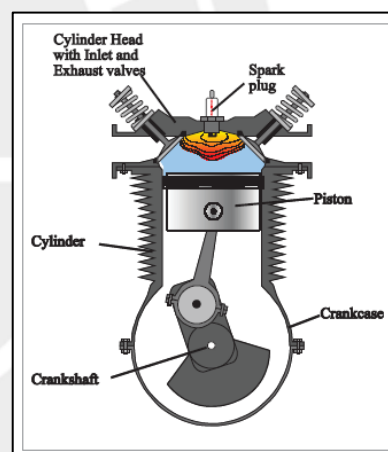
1. Boyle's Law: At constant temperature, $PV = \text{Constant}$
2. Charles's Law: At constant pressure, $\frac{V}{T} = \text{Constant}$
3. Combined Gas Laws: $\frac{PV}{T} = \text{Constant}$, $PV = nRT$

Internal Combustion Engine – Piston engine

- Piston engine is a reciprocating engine working on the principle of Otto cycle (the cycle was invented by Nikolaus Otto).
- There are 4 strokes in 1 cycle – Intake, Compression, Power and Exhaust, with only 1 useful stroke (Power stroke).
- It converts Heat energy to Mechanical energy and produces thrust by rotating the propeller.



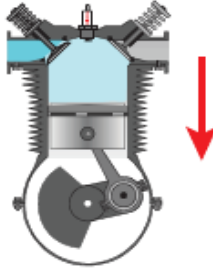
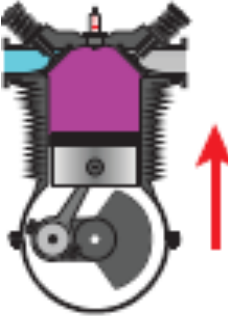
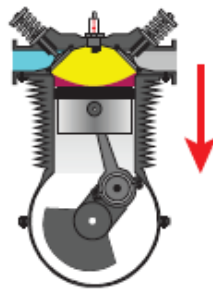
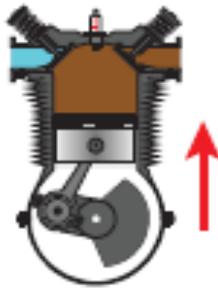
Piston – cylinder arrangement



General Parts

- **Top Dead Center (TDC):** The top most position of the piston is known as TDC.
- **Bottom Dead Center (BDC):** The bottom most position of the piston is known as BDC.
- **Stroke:** Movement of piston from TDC to BDC or from BDC to TDC is known as stroke.
- 1 Stroke = 180° rotation of crankshaft
- 2 Strokes = 1 complete revolution of crankshaft (360° rotation)
- 4 strokes = 1 Complete cycle = 720° rotation of crankshaft

4 Strokes of a piston engine:

	1. Intake - Intake valve opens, Air mass comes in, Piston moves from TDC to BDC.
	2. Compression – Both valves remain closed, Air mass gets compressed, Piston moves from BDC to TDC.
	Combustion: Both valves are closed and piston is at TDC. Electric spark is given by spark plug to ignite the mixture. Charge temperature and pressure increases rapidly during combustion. The VOLUME IS UNCHANGED due to the stationary piston, hence the Internal combustion engine is also known as a `CONSTANT VOLUME ENGINE/Combustion'. 3. Pressure acting on piston, piston extracts energy from burned gases. Both valves remain closed, Piston moves from TDC to BDC.
	4. Exhaust valve opens, the piston moves up from BDC to TDC forcing the exhaust gas out of cylinder.

Four strokes of Otto Cycle

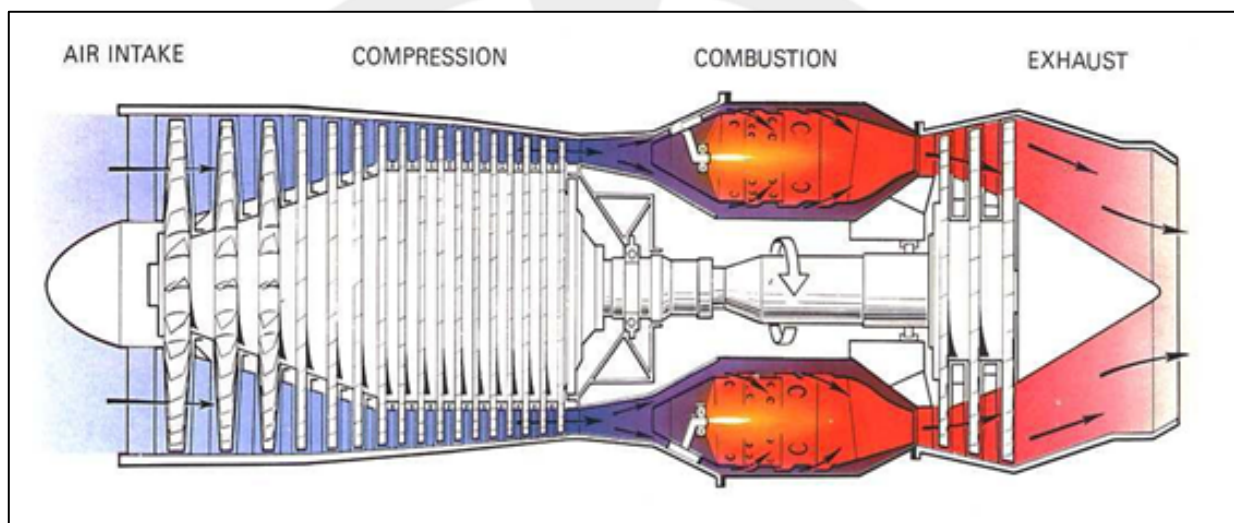
GAS TURBINE ENGINE

Jet engine operates on Newton's third Law of motion which states that for every force acting on a body, there is an equal and opposite reaction. Piston engine/propeller combination and jet engine both propel mass of air backwards and generates a reaction in opposite direction driving the aircraft forwards.

Thrust Force = Mass X Acceleration = $m \cdot a$

Working Cycle:

Gas turbine engine works on the Brayton cycle which is very similar to Otto Cycle with four stages namely, induction, compression, combustion and exhaust.



The Brayton Cycle

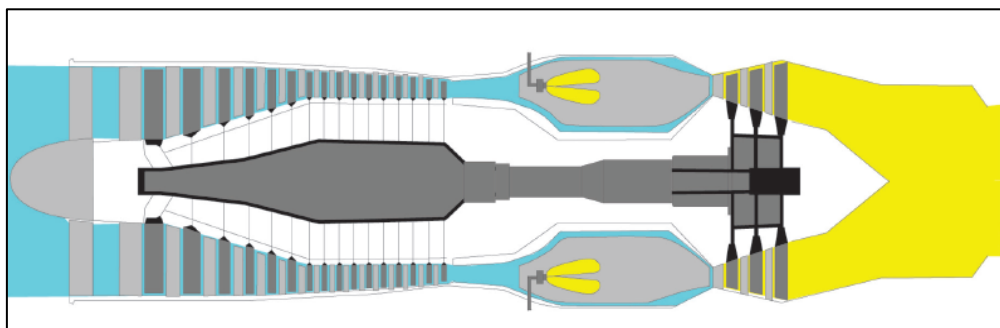
In Jet engine, Combustion theoretically occurs at constant pressure whereas in piston engine it occurs at constant volume. Power is developed in turbine of the engine. Piston engine produces intermittent power (only 1 out of 4 is power producing stroke) while gas turbine is a continuous power producing engine. That's why gas turbine has a greater power/weight ratio than piston engine.

Air at atmospheric pressure enters the engine and is compressed in compressor. Fuel is added to incoming air and burnt, in theory at a constant pressure. Gas generated through combustion expands in the turbine where energy in gas stream is converted to mechanical energy by turbine.

Compressor delivery air (pressure of air when it leaves final stage of compressor and before entering the combustion chamber) is the **highest pressure air in the engine**.

Types of Jet Engines:

1. Pure straight Turbojet Engine:

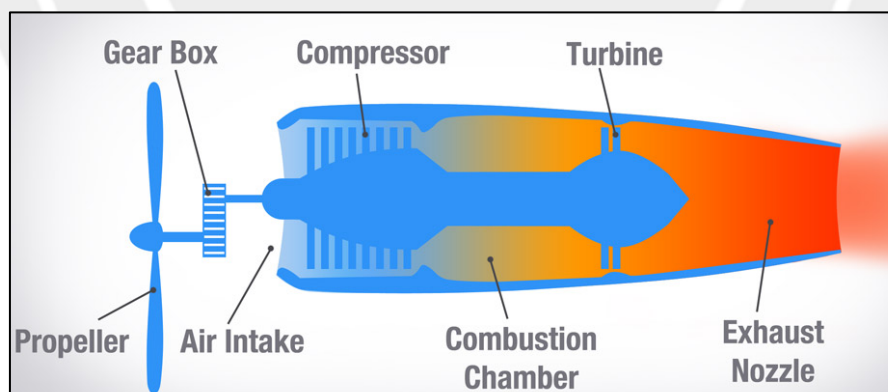


A single spool axial flow compressor turbojet engine

- When a compressor and turbine are joined on one shaft, the unit is called a spool. It was used earlier in fighter aircraft.
- Energy required to drive the compressor is extracted from the gases as they pass through turbine, remaining energy is extracted to produce thrust.

Turbojet energy = Drive Compressor + Jet Thrust

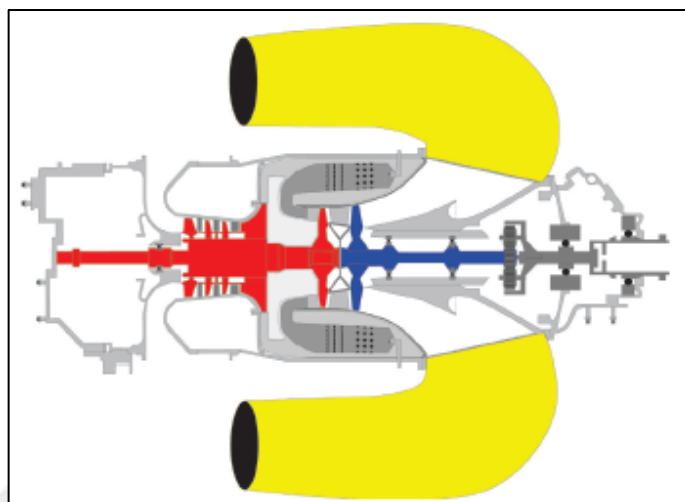
2. Turboprop Engine:



- In turboprop engine, almost all the energy in the gas stream is converted into mechanical energy.
- The energy in a turboprop is used to drive compressor, propeller and only a small amount of 'jet thrust' is available from the exhaust system.
- Most efficient engine in low speed range.

Turboprop energy = Drive compressor + Drive propeller + Jet thrust (only a small portion)

3. Turboshaft Engine:

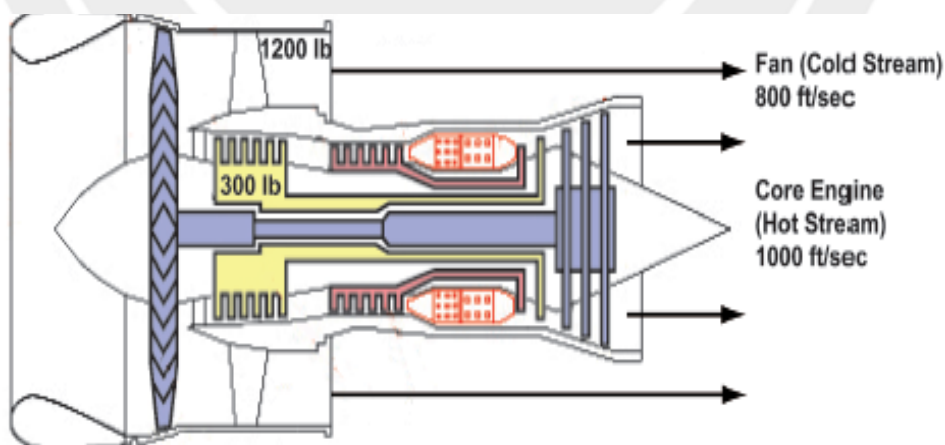


A single spool turboshaft engine with reverse flow combustion chamber system

- It can be thought of a turboprop engine with propeller replaced by a shaft.
- Turbo shaft can be used to drive helicopter rotors and are normally used as auxiliary power unit (APU) on most modern transport aircraft.

4. Turbofan Engine:

- A fan is used in front to suck in more air mass.
- At end of fan, air flow splits into two (a portion flows through bypass duct and remainder continues through high pressure compressor) and recombines in the mixer unit of the exhaust system.



$$\text{Bypass ratio} = \frac{\text{Air mass flowing through bypass duct}}{\text{Air mass flowing through combustion chamber}} = \frac{\text{Cold Air}}{\text{Hot Air}}$$

- Bypass ratio is the ratio of mass airflow which flows through the fan-duct (bypass duct) to the mass of air which is directed through the hot core. Low bypass is in region of 1 or 2:1, whereas high bypass ratio would be around 5:1.
- The triple spool front fan turbojet engine is probably the most successful example of this type of engine.
- Thrust is almost completely dependent on bypass airflow having high mass and low velocity and hence has good propulsive efficiency.
- The air passing through high pressure compressor has large energy added in combustion chamber, and this energy is required to drive the compressors.

The rearmost, or the low pressure turbine, extracts virtually all of the energy that remains in the gas stream to drive the front fan and only residual thrust remains when the hot gases emerge from the turbine.

Turbofan thrust = Drive compressor + drive fan (LP comp) + Residual Thrust

Auto Thrust:

- Aka Auto throttle /Automatic throttle
- Allows pilot to control engine power by specifying a desired flight characteristic, rather than manually controlling fuel flow.
- Greatly reduces the pilots' workload
- Helps conserve fuel - Meters precise amount of fuel
- Extends engine life

Airbus A320 NEO (New Engine Options):

- A320 NEO : New Generation Twinjet Narrowbody Airliner
 - Incorporates Sharklets – Lesser Drag, Faster cruising speed
 - Higher Bypass Ratio (~12:1), Larger Fan - Increased thrust, Improved Fuel Efficiency
 - Increased Operational Range
 - Quieter – Reduced noise inside & outside the aircraft
 - Better pressurization, LED lighting, Greater luggage space
 - Modern seating design - Greater comfort for passengers